

Project Report: Vehicle Parking App 07

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Mindful learner striving for personal and academic growth with integrity.

1. Project Description:

This is a multi-user parking application specifically designed for 4-wheeler parking. The application manages multiple parking lots and their corresponding spots. It enables users to register, log in, and reserve parking spots, while administrators (admins) can create, update, and manage parking lots and view overall system data.

2. Frameworks:

- Flask for application code
- Jinja2 templates for HTML generation and styling
- Flask-SQLAlchemy for data management
- Matplotlib for plotting graphs

3. Process, Challenges and Solutions:

- Started with making simple html pages as per the design given in the project wireframe and slowly added css to make things look good
- Added routes for the functionalities and linked them with respective html pages
- Initially had problems designing the looks with the help of css but later took help from ChatGPT and W3school.com
- Faced problems in making the relationships of models, solved it by learning it from the sqlalchemy official website

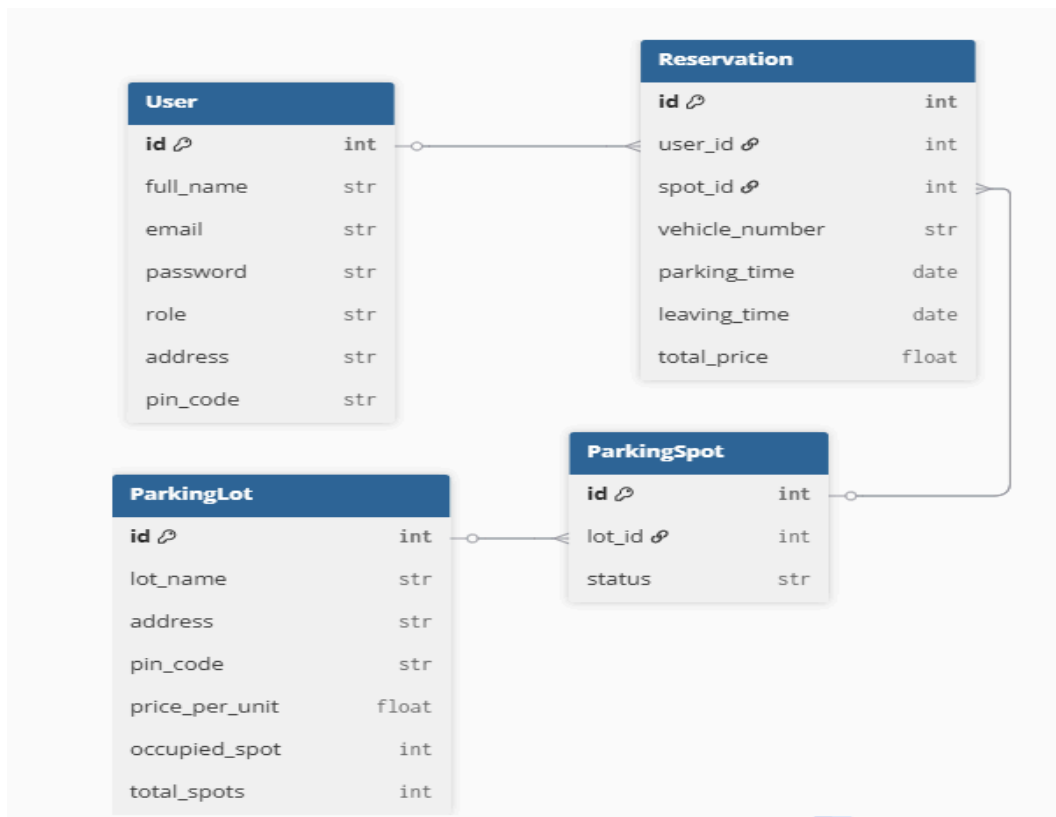
4. Functionalities:

- Users can register and log in using their email, password, and role information (user/admin role is differentiated).
- Based on their role, users are redirected to either the **User Dashboard** or **Admin Dashboard** after login using Python logic in Flask.
- **Admin can add, edit, and delete** parking lots. The number of parking spots are created automatically based on the total spot value entered while adding a lot.
- **Admin can view** the live status (Available/Occupied) of all parking spots and their details in the dashboard.
- **Admin can view all registered users** along with their personal and reservation details.
- **Users can choose a parking lot** and the system auto-allocates the first

available parking spot.

- **Users can release the spot** when they leave, updating the status of the spot and calculating total price based on parking time.
- The **parking history** of each user is stored and shown in the user dashboard.
- A **summary chart** is generated for each user based on their parking activity using `matplotlib`.
- Admin dashboard also shows **charts and analytics** summarizing lot-wise usage.

5. E-R Diagram:



6. Conclusion:

The **Vehicle Parking App - V1** successfully implements a multi-user platform for managing 4-wheeler parking systems with distinct roles for admin and users. The application enables the admin to manage parking lots and spots efficiently, while users can register, book, and release parking spots seamlessly.

7. Presentation link:

- <https://drive.google.com/file/d/1p7z5pdpCIQ6zx2zOeT3hAAo1Y6dupSGF/view?usp=sharing>

